Abdulrahman Eldeeb

Python Developer

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Summary

Computer Science graduate and Codeforces Expert focused on backend development with Python – Django, Django REST Framework, and FastAPI. Experienced in building production-ready RESTful APIs using Dockerized workflows, JWT authentication with Redis-backed revocation, Celery background tasks, automated testing, and OpenAPI documentation. Final-year graduation project (Grade A+) in on-device AI driver monitoring (YOLOv8, Face Mesh), with additional ML/DL background in CNNs, transfer learning, and XGBoost.

Professional Experience

Instructor Sep 2024 – Aug 2026

Recode Academy *Fayoum, Egypt*

- Taught multiple courses including CS50, C++ programming, algorithms and data structures, and Object-Oriented Programming (OOP) core concepts.
- Delivered structured curriculum covering problem-solving fundamentals, computational thinking, and advanced programming paradigms to diverse student cohorts.

Mentor & Coach April 2023 – Mar 2025

Fayoum ICPC Community *Fayoum, Egypt*

- Mentored 15–20 students in programming fundamentals, data structures, and algorithm techniques to prepare for the Egyptian Collegiate Programming Contest (ECPC).
- Organized programming workshops, training sessions, and coding competitions to enhance programming abilities of community members.

Backend Projects

Bookly - Book Management API | FastAPI, SQLModel, PostgreSQL, Redis, Celery Aug 2026 – Sep 2026

- Built an async book catalog API with FastAPI + SQLModel covering JWT auth with refresh rotation, **Redis blocklist** for revoked tokens, role-based access, and owner-scoped book CRUD with reviews and tags.
- Implemented **Celery background tasks** for email delivery – verification, resend-verification, and password-reset flows via FastAPI-Mail with signed time-limited tokens.
- Hardened the API with Redis **rate limiting** on auth endpoints, Alembic migrations, centralized error codes, and auto-generated Swagger/ReDoc docs.

E-Commerce REST API | Django REST Framework, PostgreSQL, Stripe, JWT, Redis Feb 2026 – Jun 2026

- Built a production-ready e-commerce API covering user auth, products, cart, orders, and **Stripe** payment processing with webhook-driven status updates.
- Applied **Service Layer & Selector Layer** patterns separating business logic from views and data access across 5 modular Django apps.
- Implemented **JWT auth** with token rotation, blacklisting, HTTP-only cookies, and split settings (base/dev/prod) with auto-generated Swagger docs.
- Added **Redis caching** with tiered TTLs, service-layer cache invalidation on writes, and environment-specific configurations.

Recipe API | Django REST Framework, PostgreSQL, Docker, OpenAPI, TDD Mar 2026 – Apr 2026

- Built a production-style Recipe Management API with user registration, token authentication, profile endpoints, and strict ownership isolation for user data.
- Implemented full CRUD for recipes, tags, and ingredients with advanced filtering (by tags/ingredients), assigned-only queries, and multipart image upload endpoints.
- Applied test-driven development with comprehensive unit tests, health checks, DB readiness workflow, and Docker Compose setup for consistent local and deployment environments.

AI Projects

Ala Mahlak – Driver Monitoring System | Graduation Project – Grade A+ May 2026

- Built an on-device real-time Driver Monitoring System processing the front-facing camera at ∼15 FPS – entirely offline with zero cloud dependency.
- Implemented **Face Mesh** (468 landmarks) for Eye/Mouth Aspect Ratio (EAR/MAR) and gaze estimation, plus pure-Dart **head pose estimation** (pitch/yaw/roll), and **YOLOv8nano** (ONNX, 4–5 MB) cabin object detection with NMS (IoU 0.45).
- Designed a **priority-based multi-signal fusion engine** (drowsiness score + YOLO + head pose + gaze) with **PERCLOS** rolling-window analysis, blink detection, state debouncing, and self-calibration – classifying 7 distraction states with audio/haptic alerts.

Deep Learning Image Classification | PyTorch, Keras, ResNet Mar 2025 – Sep 2025

- Built facial emotion recognition on FER-2013 comparing a custom CNN from scratch vs. transfer learning with **ResNet50V2** – EDA, class-balanced training, and confusion-matrix evaluation.
- Classified natural scenes (Intel dataset, 6 categories) with **AlexNet and ResNet18** in PyTorch – full pipeline: preprocessing, training, hyperparameter tuning, evaluation, and inference.

Open Source

- **Open-source contributor** – merged pull requests into community skill libraries for coding-agent delegation.
- Refactored seven duplicated relay event scanners into one parity-gated implementation, fixing an unbounded memory-growth defect and adding 21 recovery test cases (PR #39).
- Shipped implementer delegation skills for the Cline and GitHub Copilot CLIs – Node-built-ins-only relays with brief-on-stdin transport and plan/read-only safety (PR #59, PR #87).

Education

Fayoum University Fayoum, Egypt

Bachelor of Computer Science and Artificial Intelligence *Oct 2022 – Jun 2026*

- Relevant Coursework: Object-Oriented Programming, Data Structures, Algorithms Design & Analysis, Operating Systems, Database Management Systems, Computer Architecture

Technical Skills

Languages	Python, C++, SQL, HTML/CSS, JavaScript, Dart
Backend	Django, Django REST Framework, Django ORM, FastAPI, SQLModel, Pydantic, Celery, Class-Based Views, ViewSets, Routers
Databases	PostgreSQL, SQLite, MySQL, Redis
APIs & Auth	RESTful APIs, Token Authentication, JWT (SimpleJWT, PyJWT), Redis Blocklist, Permissions, Throttling, Rate Limiting, Stripe Integration, Webhooks, Swagger/OpenAPI
Architecture	MVT, Service Layer, Selector Layer, Modular App Design, RBAC, Async Python, Background Tasks, Clean Architecture, Test-Driven Development (TDD)
Computer Vision	OpenCV, MediaPipe, Google ML Kit, YOLOv8, ONNX Runtime, Face Mesh, Head Pose Estimation, EAR/MAR, PERCLOS, Gaze Estimation
Deep Learning	PyTorch, TensorFlow/Keras, CNNs, ResNet, AlexNet, Transfer Learning, DNN Module
Machine Learning	Scikit-Learn, XGBoost, Regression, Classification, Clustering, Ensembles, Feature Engineering
Data & Analytics	Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Power BI
Tools	Git, GitHub, GitHub Actions (CI/CD), Docker, Docker Compose, Alembic, Uvicorn, Flake8, Postman, pip, virtualenv, django-filter, Streamlit
AI-Assisted Dev	Coding agents (opencode, Claude Code), Model Context Protocol (MCP), multi-agent systems, agent orchestration, skill/tool development for AI workflows
Other	Web Scraping (Beautiful Soup, Selenium), Jupyter Notebooks, Google Colab, Flutter, BLoC

Competitive Programming

Codeforces Expert – Demonstrated strong algorithmic problem-solving skills

Ranked 17th Place in 2023 ECPC Qualifications (out of 280 teams)

Ranked 90th Place in the 2023 ECPC Final (out of 280 teams)

Certifications

- **Build REST APIs with Django REST Framework and Python** – Shubham Sarda (Udemy), 13 hours. Focus: serializers, APIViews/Generic Views, ViewSets & Routers, permissions, auth (Basic/Token/JWT), throttling, filtering, pagination, and API testing with Postman.
- **Python Django - The Practical Guide** – Maximilian Schwarzmuller (Academind/Udemy), 23 hours. Focus: Django fundamentals, templates/static files, ORM relationships, forms, sessions, class-based views, file uploads, and deployment workflow.
- **Build a Backend REST API with Python & Django - Advanced** – Mark Winterbottom & Brooke Rutherford (Udemy), 15 hours. Focus: Docker, TDD, GitHub Actions CI/CD, Flake8 linting, custom user auth, image upload APIs, filtering/search, and production-oriented project structure.
- **Machine Learning Specialization** – Andrew Ng, Stanford / DeepLearning.AI. Supervised learning, advanced algorithms (ensembles, clustering, recommender systems), ML engineering (bias/variance, error analysis, sklearn workflows).
- **IBM Data Science Professional Certificate** – DEPI. Python for data science, SQL, data analysis/visualization (Pandas, Matplotlib, Seaborn), ML with Scikit-Learn.
- **Deep Learning for Computer Vision + Computer Vision Applications** – coursat.ai. CNNs, backpropagation, training pipelines; image classification with AlexNet/ResNet, facial emotion recognition, transfer learning.

Soft Skills

- Technical Training
- Curriculum Development
- Team Leadership
- Problem-Solving
- Coaching & Feedback
- Communication